

`In[ ]:= Integrate[ $\frac{v^4}{v^3 + vc^3}$ , {v, 0, v0}, Assumptions -> {v0 > vc, vc > 0}]`
 [积分] [假设]

`Out[ ]:=`

$$\frac{v_0^2}{2} - \frac{1}{18} vc^2 \left(\sqrt{3} \pi + 6 \sqrt{3} \operatorname{ArcCot} \left[\frac{\sqrt{3} vc}{2 v_0 - vc} \right] - 6 \operatorname{Log}[v_0 + vc] + 3 \operatorname{Log}[v_0^2 - v_0 vc + vc^2] \right)$$

`In[ ]:= FullSimplify[%]`
 [完全简化]

`Out[ ]:=`

$$\frac{v_0^2}{2} - \frac{1}{18} vc^2 \left(\sqrt{3} \pi + 6 \sqrt{3} \operatorname{ArcCot} \left[\frac{\sqrt{3} vc}{2 v_0 - vc} \right] - 6 \operatorname{Log}[v_0 + vc] + 3 \operatorname{Log}[v_0^2 - v_0 vc + vc^2] \right)$$